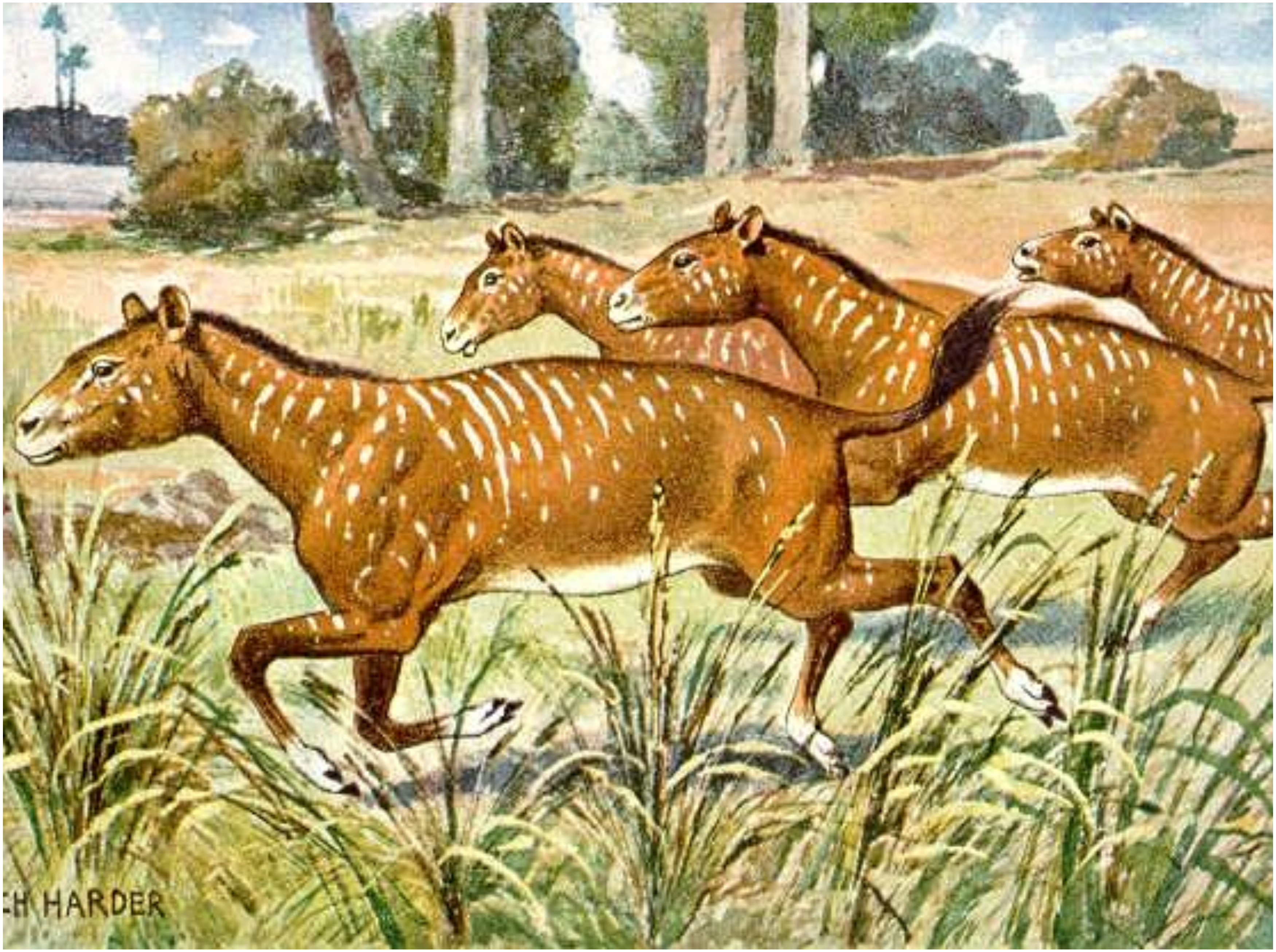
Straight from the Horse's Mouth: Survival of the Toothiest - Epoch Card

Eocene Flora
(55.8 - 33.9 MA)

At the beginning of the Eocene, the high temperatures and warm oceans created a moist, balmy environment, with **forests** spreading throughout the Earth from pole to pole. Apart from the driest deserts, scientists hypothesize the Earth was entirely covered in forests. Polar forests were quite extensive. Fossils and preserved remains of trees such as swamp cypress and dawn redwood from the Eocene have been found in the Arctic. Fossils of subtropical and even tropical trees and plants from the Eocene have also been found in Greenland and Alaska. Tropical rainforests grew as far north as northern North America and Europe. Palm trees were growing as far north as Alaska and northern Europe during the early Eocene, although they became less abundant as the climate cooled. Cooling began mid-epoch, and by the end of the Eocene continental interiors had begun to dry out, with forests thinning considerably in some areas. The cooling also brought seasonal changes. Deciduous trees, better able to cope with large temperature changes, began to overtake evergreen tropical species. By the end of the epoch, deciduous forests covered large parts of the northern continents, including North America, Eurasia and the Arctic, and rainforests only remained in equatorial South America, Africa, India and Australia.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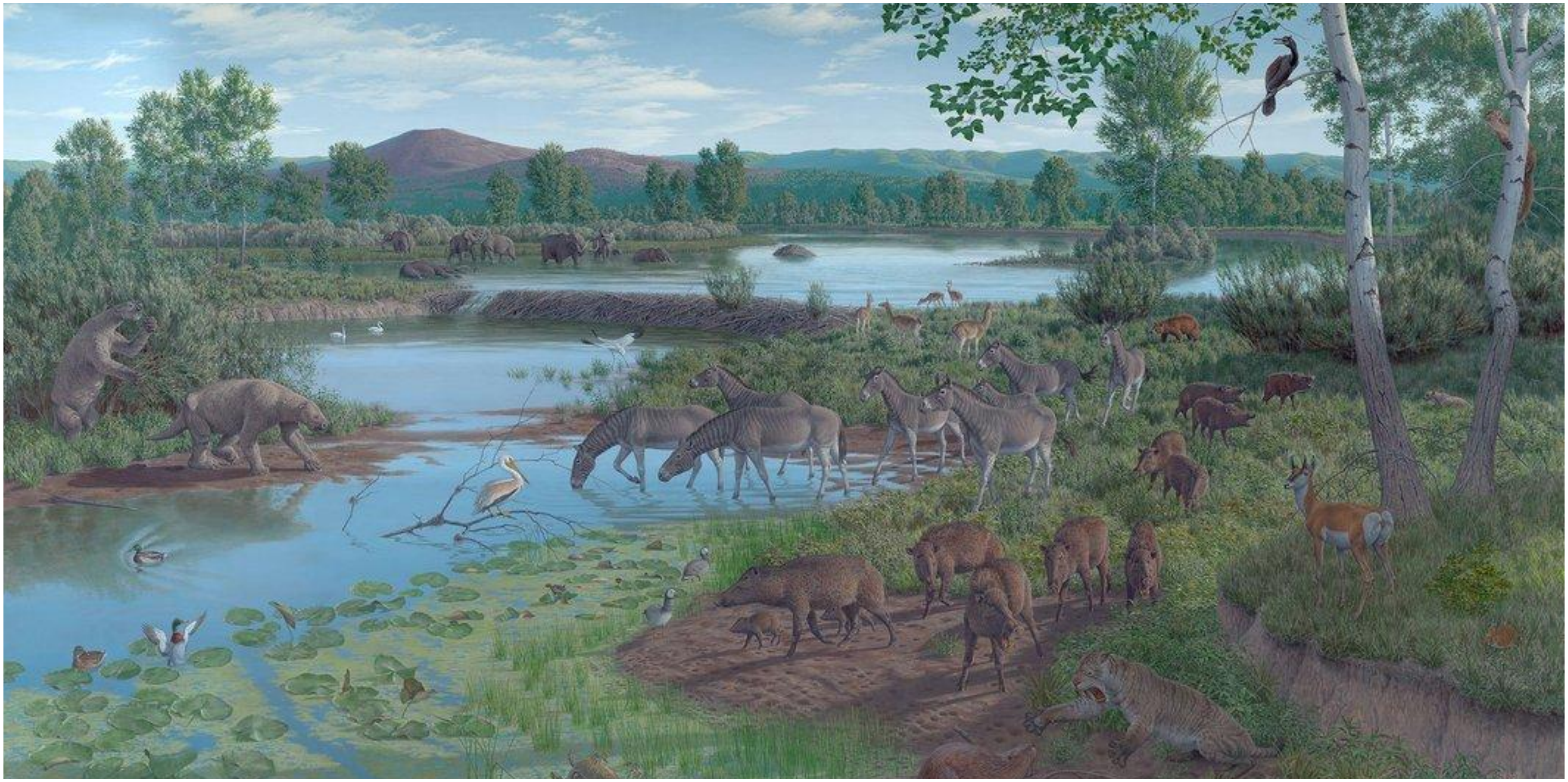
Oligocene Flora
(33.9 - 23 MA)

Angiosperms continued their expansion throughout the world as temperate deciduous forests replaced tropical and sub-tropical forests. **Grasslands began to expand** and forests (especially tropical ones) began to shrink. Grasses, a product of the cooler, drier climate, became one of the most important groups of organisms on the planet. As they spread extensively over several million years, they fed herds of grazing mammals, sheltered smaller animals and birds, and stabilized soil, which in turn reduced erosion. They are high-fiber, low-protein plants and must be eaten in large quantities to provide adequate nutrition. Because they contain tiny silica fragments, though, they are tough to chew and wear down animal teeth. Grasses, which grow throughout the blade, are adapted to recover quickly after their tips are grazed. However, even at the end of the epoch, grass was not as common as one would observe in a modern savannah today.

Miocene Flora
(23 - 5.3 MA)

Plant studies of the Miocene have focused primarily on spores and pollen. Such studies show that by the end of the Miocene 95% of modern seed plant families existed. A mid-Miocene warming, followed by a cooling is considered responsible for the retreat of tropical ecosystems, and increased seasonality. Prior to this time all plants were classified as C3 because the CO2 is first incorporated into a 3-carbon compound. In C3 the enzyme rubisco is involved in the uptake of CO2 . While C3 plants function very well in cool, moist conditions as the temperature warmed, a new photosynthetic pathway evolved. This new pathway was called C4 because the CO2 is first incorporated into a 4-carbon compound. C4 plants photosynthesize faster than C3 plants under high light intensity and high temperatures. Although the C4 pathway is more complicated (requires more enzymes and specialized plant anatomy), C4 plants do not need to keep stomata open as much and lose less water to transpiration (evaporation of water through stomata). The C4 pathway also causes the uptake of more silica (a glass-like material) that resulted in the diversification of modern grasses and sedges. The evolution of gritty, fibrous, fire-tolerant grasses led to a **major expansion of grass-grazer ecosystems**, with roaming herds of large, swift grazers pursued by predators across broad sweeps of open grasslands, displacing desert, woodland, and browsers.



Pliocene Flora
(5.3 - 2.6 MA)

The change to a cooler, dry, seasonal climate had considerable impacts on Pliocene vegetation, reducing tropical species worldwide. Tundra covered much of the north, and grasslands spread on all continents (except Antarctica). Tropical forests were limited to a small band around the equator, and in addition to dry savannahs, deserts appeared in Asia and Africa. In the higher latitudes, cool-weather plants evolve. In lower latitudes, **grasslands** are marked by fewer and fewer trees. These habitats offer limited food sources for animals and support less diversity.

Pleistocene Flora
(2.6 - 0.1 MA)

During the Pleistocene, glaciers repeatedly advance from the Arctic north over Europe and North America, then retreat. The first major glacial flow occurs about 1.6 Ma. Ice, up to a mile thick in places, spreads from Greenland over the Arctic Sea into northern Europe and Canada. As the ice advances, temperatures ahead of the flow drop significantly. The temperature change has a profound impact on life. Mammoths, rhinos, bison, reindeer, and musk oxen all evolve to have warm, woolly coats to protect them from frigid conditions. These new mammals feed on the small bushes and **hardy grasses** that tolerate cold as they follow the moving line of glaciers.